

say, for instance, 8-bit, 10-bit, 12-bit or 24-bit.

The goal of an ADC is always to detect even a very small range of signals. So, for example, it is possible to resolve only a signal bound by the quantisation error. In lower numbers, it may be extremely difficult to detect that signal because it will all be represented by the same digital code. But when you increase the number of levels, you can detect even smaller signals.

The ratio of the smallest resolvable signal to the largest resolvable signal is defined by the maximum achievable dynamic range. With higher resolution, you get better dynamic range and capability to even detect small signals.

For example, in a 2.5V full scale ADC, even a 95-microvolt signal can be detected with an 18-bit ADC, and a 12-bit ADC can detect only around 610 microvolts of signal.

And for some analogue inputs, one must define polarity—single ended, unipolar, or differential. This should be considered if you are looking for high precision systems and want to ensure less components.

Picking the right architecture

Each ADC architecture comes with its own pros and cons. Being aware of these will help you select the right type of architecture for your requirements. Some of the well-known architectures are flash, successive approximation (SAR), delta-sigma, and pipelined.

SAR ADC is the predominantly-used architecture, tailored for instrumentation, industrial control, and data type of applications. Its key features involve the ability to control conversion process, start pulse, reset logic, and comparator efficiency and other notable advantages over low power, small die area, and cost. So, this type of architecture is very well used in 8-bit, 10-bit, and 12-bit ADCs.

But the flip side is, the number

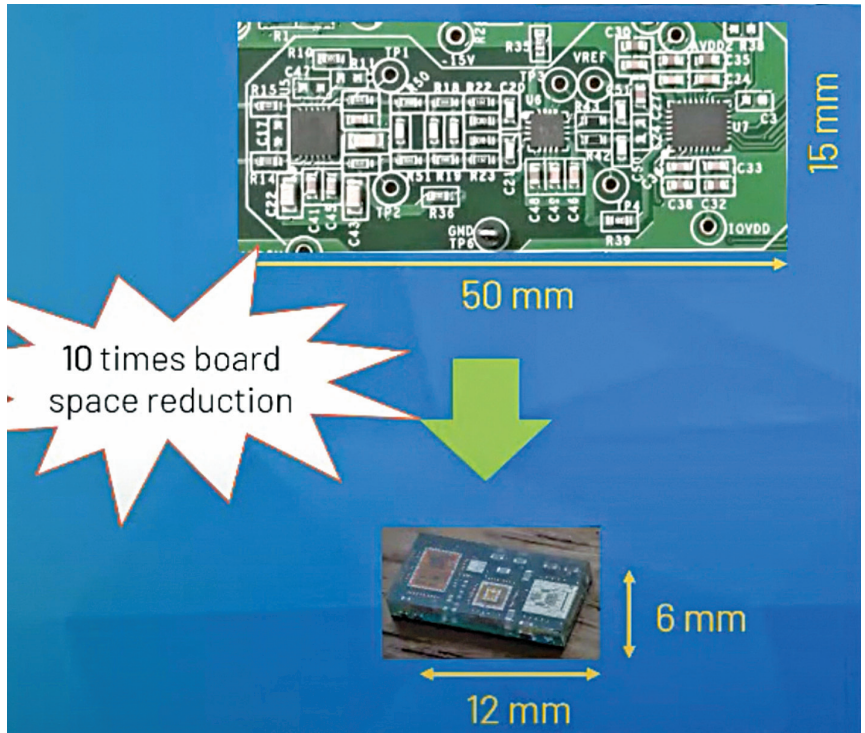


Fig. 3: MM module data acquisition system (Source: Analog Devices)

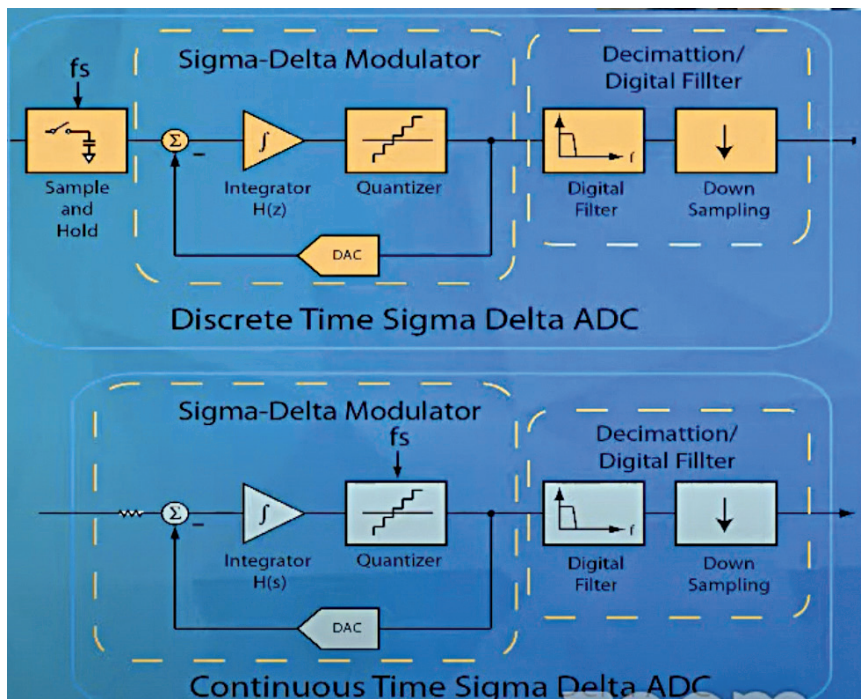


Fig. 4: Discrete time signal delta ADC vs continuous time sigma delta ADC

of resolutions and comparisons will significantly increase conversion optimisation time. Hence, this architecture is feasible when the resolution parameter is not very high.

The other architecture is sigma delta ADC, which is used for high-precision applications requiring higher resolution. Concepts of over-sampling and noise shaping, wherein